Project SocMag: University Society Discovery, Dual-Track Recruitment & Digital Event Ecosystem

Project Proposal for UCS503P
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1 Higher-order goal

This project aims to modernize and centralize student extracurricular discovery, recruitment, and event engagement across university societies (for example: CCS, OWASP, ACM, IEEE, MUDRA, FAPS, Pratibha, EDC). While organizational management spans wide institutional domains, the immediate engineering objective is to translate *Maximum Likelihood Estimation (MLE) applicant matching, proctored dual-track evaluation, real-time audition slot scheduling, centralized event feeds, and digital event management* into a deployable, high-throughput software ecosystem.

2 Time-to-value

- We will deliver meaningful increments quickly by prototyping the interactive society showcase, MLE-based Compatibility Engine, and applicant tracker dashboard within a short iteration window.
- We will prioritize fast feedback over perfection by using weekly agile sprints backed by CI automation to validate judge rubric panels and proctored technical assessment workflows early.
- Success is defined by deploying a functional showcase, dual-track recruitment flow (Technical and Cultural), and digital event management with feedback integration in Sprint 1, then refining statistical persona modeling and anti-cheat algorithms based on pilot feedback.

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3 Problem Statement

During campus recruitment drives and major event seasons, first-year students blindly apply to dozens of societies without knowing cultural or domain fit, while societies struggle with unorganized recruitment pipelines and fragmented event management. Core committees face severe operational friction across discovery, assessment, and event handling. Common pain points include:

- **Unfiltered Applicant Floods and Misalignment:** Students apply blindly without clear visibility into society domains (Technical, Cultural, Literary, Entrepreneurial) or skill compatibility.
- **Recruitment Bottlenecks across Tracks:** Technical societies (CCS, OWASP, ACM, IEEE) lack integrated, proctored testing environments with anti-cheat logging, while cultural and non-technical societies (MUDRA, FAPS, Pratibha, EDC) suffer from chaotic physical audition queues and media handling.
- **Subjective and Unstandardized Grading:** Interview evaluations are frequently fragmented across unstructured spreadsheets without dynamic rubric aggregation for panel judges.
- **Campus Event and Feedback Friction:** Event announcements and registrations are decentralized, tracking attendance across disjointed platforms is inefficient, and post-event feedback collection is fragmented.

Why this matters now (contextual relevance): In a dense institutional ecosystem, integrating intelligent MLE persona matching, dynamic audition slotting, proctored domain testing, and centralized event feeds eliminates administrative burnout and maximizes student engagement.

4 Proposed Solution

4.1 Overview

Build a comprehensive web platform titled **Project SocMag** featuring three integrated core modules:

- **Module 1: Discovery, Onboarding and Statistical Matching:** Interactive society profiles, dynamic feature vector extraction paired with a Maximum Likelihood Estimation (MLE) engine to compute a statistically sound *Compatibility Score (%)* against a society's "ideal persona," and a unified application tracker dashboard (Applied to Test or Audition Scheduled to Interviewing to Selected).
- **Module 2: Dual-Track Recruitment Engine:**
  - **Track A (Technical Societies - CCS, OWASP, ACM, IEEE):** In-app assessment module featuring automated MCQ and coding tests for Data Structures and Algorithms (DSA), Web Development, and Cybersecurity domains, coupled with a time-bound proctored test window featuring tab-switch detection and anti-cheat event logging.
  - **Track B (Non-Technical and Cultural Societies - MUDRA, FAPS, Pratibha, EDC):** Direct upload interface for design portfolios (FAPS), audition video clips (MUDRA/Pratibha), or business pitches (EDC); a real-time audition slot booking system for in-person interviews to eliminate physical lines; and a private Recruiter

Rubric and Judge Panel UI allowing core committee members to grade applicants on explicit parameters (for example: Confidence, Technique, Creativity) with automated aggregate score calculations.

- **Module 3: Society Events and Digital Management:** Centralizes event announcements, registrations, and attendance tracking across campus through a Unified Campus Event Feed (a categorized timeline of upcoming hackathons, workshops, cultural nights, and guest lectures across all societies), event registration workflows, and an integrated post-event Feedback Pipeline.

4.2 Core workflow

1. Freshers complete the onboarding profile; the MLE engine evaluates their feature vector against ideal society personas to display dynamic Compatibility Scores across profiles.
2. Applicants submit applications through the central dashboard:
   - **Track A applicants** enter the time-bound test environment to complete auto-graded DSA, Web Dev, or Cybersecurity tests under proctored anti-cheat monitoring.
   - **Track B applicants** upload portfolios/videos/pitches and reserve dynamic real-time physical audition time slots.
3. Committee judges evaluate Track A auto-scores or grade Track B candidates using private rubric boards with dynamic parameter aggregation (Confidence, Technique, Creativity).
4. Students explore the Unified Campus Event Feed, register for events, and complete post-event forms through the feedback portal.

4.3 Operational constraints for an educational campus setting

- Maintain sub-second responsiveness during peak assessment submission and event registration hours.
- Decouple code execution, anti-cheat detection, and statistical MLE matching into dedicated backend microservices.
- Ensure dynamic client-side rendering for real-time audition slot booking synchronization.

5 Solution Approach

This engineering project emphasizes robust software architectural design, proctored assessment pipelines, statistical likelihood estimation, and relational database normalization.

5.1 Statistical and Mathematical Engine

The platform relies on rigorous mathematical and algorithmic models:

- **MLE Compatibility Engine:** Models the "ideal persona" of each society as a parametrized probability distribution. The engine maps applicant attributes, domain experience, and personality traits into a feature vector. It computes the likelihood function using Maximum Likelihood Estimation (MLE) to generate a normalized, statistically sound Compatibility Score ($0 - 100\%$).
- **Track A Auto-Grading and Proctoring Engine:** Executes time-bound test evaluations for coding and MCQ formats across DSA, Web Dev, and Cybersecurity. Computes dynamic scores while logging tab-switch frequency and focus-loss duration for anti-cheat scoring.

- **Track B Slot Scheduling and Rubric Aggregation:** Models candidate physical throughput to assign non-overlapping audition slots dynamically. Aggregates multi-judge scores across weighted parameters for Confidence, Technique, and Creativity.

5.2 Web and Data Architecture

A scalable decoupled multi-tier architecture:

- **Frontend:** Next.js (React) with Framer Motion transitions, Chart.js radar charts for rubric visualization, and responsive user interface components.
- **Backend API:** Node.js (Express) handling RESTful endpoints, audition slot reservations, applicant state transitions, and event feeds.
- **Execution and Math Microservice:** Python microservice executing code auto-grading, tab-switch anomaly logging, MLE Compatibility calculations, and automated PDF report generation.
- **Database:** PostgreSQL relational schema strictly normalized to BCNF (`Users`, `Societies`, `Applications`, `Test_Submissions`, `Media_Uploads`, `Slot_Bookings`, `Rubric_Grades`, `Event_Registratio` `Feedback_Responses`).

5.3 SE Lifecycle, Integration, and CI/CD

Software engineering practices include:

- **Agile/XP Sprints:** Scrum velocity tracking with structured User Stories across all three modules.
- **Formal Design Artifacts:** Data Flow Diagrams (DFDs), Entity-Relationship (ER) diagrams, and BCNF schema validation.
- **Testing and CI/CD Automation:** Controller-service separation, white-box testing for auto-grading and MLE likelihood logic, and automated GitHub Actions deployment pipelines.

6 Evaluation Criterion

We measure project success using quantitative metrics derived from campus recruitment and event operational data.

6.1 Primary evaluation metric

Recruitment and Event Operational Throughput: Total latency reduced across audition processing and event registration execution.

- Target: Reduction of physical audition queue wait times by 40% or greater through real-time slot reservation.
- Attribution: Database microsecond timestamps logging candidate slot arrivals and judge evaluation submissions.

6.2 Secondary evaluation metrics

- **Proctoring Integrity:** Low false-positive rates in tab-switch and anti-cheat event detection during Track A technical assessments.
- **MLE Match Relevance:** High statistical correlation between the computed MLE Compatibility Score and applicant conversion rate into active members.
- **Judge Rubric Reliability:** High inter-rater consistency across aggregated multi-judge scores in Track B auditions.

6.3 Pilot validation plan

- Deploy pilot across 100+ first-year students and 8 diverse campus societies (Technical: CCS, OWASP, ACM, IEEE; Non-Technical/Cultural: MUDRA, FAPS, Pratibha, EDC).
- Measure recruitment throughput, event registration performance, media upload handling, and candidate satisfaction against traditional manual drives.

7 Scalability

The solution scales both theoretically and operationally to handle institute-wide recruitment drives and event engagement.

1. **Scalable (theoretically):** Supports high concurrent loads during campus-wide drives and event registrations by:
   - decoupling heavy code execution, media processing, and MLE computation into isolated asynchronous microservices,
   - leveraging BCNF database indexing for low-latency query performance during slot reservations,
   - enforcing stateless JWT authentication across RESTful API endpoints.
2. **Operationally deployable (practically):**
   - containerized microservices managed using Docker,
   - cloud PostgreSQL database hosting,
   - automated continuous deployment pipelines using GitHub Actions.

8 Engine availability heuristic

Mature development stacks exist to rapidly build and deploy this system:

- Next.js and Express (Node.js) for high-performance visual dashboards, event feeds, and REST API management.
- Python (SciPy, NumPy, ReportLab) runtime environment for MLE statistical modeling, code auto-grading, proctoring logging, and dynamic report generation.
- PostgreSQL relational DBMS with BCNF-compliant schema execution.
- Chart.js for visual rubric analytics.
- GitHub Actions for automated unit testing and staging builds.

9 Project scope and deliverables

9.1 Initial deliverable

- Interactive society showcase profiles and MLE-based persona onboarding module with Compatibility Scoring.
- Initial PostgreSQL BCNF relational schema deployment for users, societies, and applications.
- Centralized student application tracking dashboard (Applied to Selected).
- Automated CI pipeline verifying core API endpoints and MLE math calculations.

9.2 Subsequent deliverables

- Track A in-app assessment module (MCQ and coding for DSA, Web Dev, Cybersecurity) with proctoring tab-switch logging.
- Track B direct media upload interface (portfolios/videos/pitches), real-time audition slot booking system, and judge rubric board.
- Module 3 Unified Campus Event Feed, event registration module, and post-event Feedback Pipeline.

10 Risks and mitigations

- **Proctoring Anti-Cheat False Positives:** Implement configurable tab-switch delay thresholds and log full event sequences rather than instant test termination.
- **Large Media Upload Payload Spikes (Track B):** Enforce client-side image/video compression and direct-to-cloud signed URL uploads to reduce server memory pressure.
- **Subjective Evaluation Bias in Auditions:** Enforce standardized multi-parameter rubric scoring (Confidence, Technique, Creativity) with dynamic inter-judge normalization.

11 Summary

Project SocMag presents a unified campus ecosystem designed to streamline society discovery, dual-track recruitment, and digital event management. By combining Maximum Likelihood Estimation (MLE) persona matching, proctored Track A technical assessments (DSA, Web Dev, Cybersecurity), Track B portfolio media handling, real-time audition slot booking, and judge rubric UI with centralized event feeds, event registration, and feedback pipelines, the platform replaces manual recruitment chaos with seamless, data-driven automation. Developed using modern agile practices, BCNF database modeling, and CI/CD automation, SocMag delivers high operational efficiency for students, recruiters, and event organizers alike.